

Sheet :- 7(Dictionary)

Q1.Frequency of Number

Given N array elements . Find the frequency of element in array. (Solve it by using dictionary.)

Input 1:

A = [2, 6, 3, 8, 2, 8, 2, 3, 8, 8]

B = 2

Output 1:

3

Q2.Find the first non-repeating element

Given N array elements . Find the first non-repeating element

Input 1:

A = [1, 2, 3, 1, 2, 5]

Output 1:

3

Q3. Check Palindrome

Given a string A consisting of lowercase characters.

Check if characters of the given string can be rearranged to form a palindrome.

Print 1 if it is possible to rearrange the characters of the string A such that it becomes a palindrome else print 0.

Example Input

Input 1:

A = "abcde"

Input 2:

A = "abbaee"

Example Output

Output 1:

0

Output 2:

1

Example Explanation

Explanation 1:

No possible rearrangement to make the string palindrome.

Explanation 2:

Given string "abbaee" can be rearranged to "aebbea" to form a palindrome.

Q4. First Repeating element

Given an integer array A of size N, find the first repeating element in it.

We need to find the element that occurs more than once and whose index of the first occurrence is the smallest.

If there is no repeating element, print -1.

Example Input

Input 1:

A = [10, 5, 3, 4, 3, 5, 6]

Input 2:

A = [6, 10, 5, 4, 9, 120]

Example Output

Output 1:

5

Output 2:

-1

Example Explanation

Explanation 1:

5 is the first element that repeats

Explanation 2:

There is no repeating element, output -1

Q5. Merge two Python dictionaries into one

Input 1:

```
dict1 = {'Ten': 10, 'Twenty': 20, 'Thirty': 30}
```

```
dict2 = {'Thirty': 30, 'Fourty': 40, 'Fifty': 50}
```

Output 1:

```
{'Ten': 10, 'Twenty': 20, 'Thirty': 30, 'Fourty': 40, 'Fifty': 50}
```

Q6.Check if a value exists in a dictionary

Input 1:

```
dict1 = {'a': 100, 'b': 400, 'c': 300}
```

Output 1:

400 present in a dict

Q7.Iterate through the keys in the dictionary.

Input 1:

```
person = {"name": "abc", "age": 25}
```

Output 1:

name

age

Q8.Iterate through the values in the dictionary.

Input 1:

```
person = {"name": "abc", "age": 25}
```

Output 1:

abc

25

Q9 .Iterate through both keys and values in the dictionary.

Input 1:

```
person = {"name": "abc", "age": 25}
```

Output 1:

name abc

age 25

Q10.Remove all elements from a dictionary.

Input 1:

```
person = {"name": "abc", "age": 25}
```

Output 1:

```
{}
```

Q11. Get a list of all keys in a dictionary.

Input 1:

```
person = {"name": "abc", "age": 25}
```

Output 1:

```
['name', 'age']
```

Q12. Get a list of all values in a dictionary.

Input 1:

```
person = {"name": "abc", "age": 25}
```

Output 1:

```
['abc', 25]
```

Q13. Generate a dictionary

Write a Python script to generate and print a dictionary that contains a number (between 1 and n) in the form (x, x*x).

Input 1:

```
N=5
```

Output 1:

```
{1: 1, 2: 4, 3: 9, 4: 16, 5: 25}
```

Q14. Create a list of tuples from the dictionary

Input 1:

```
dict1 = { 1: 'a', 2: 'b', 3: 'c' }
```

Output 1:

```
[(1, 'a'), (2, 'b'), (3, 'c')]
```

Q15. Write a code to sort dictionaries using a key.

Input 1:

```
{2: 'Apple', 1: 'Mango', 3: 'Orange', 4: 'Banana'}
```

Output 1:

1 Mango

2 Apple

3 Orange

4 Banana

Q16. Common Elements

Given two integer arrays, A and B of size N and M, respectively. Your task is to find all the common elements in both the array.

NOTE:

- Each element in the result should appear as many times as it appears in both arrays.
- The result can be in any order.

Example Input

Input 1:

A = [1, 2, 2, 1]

B = [2, 3, 1, 2]

Input 2:

A = [2, 1, 4, 10]

B = [3, 6, 2, 10, 10]

Example Output

Output 1:

[1, 2, 2]

Output 2:

[2, 10]

Example Explanation

Explanation 1:

Elements (1, 2, 2) appears in both the array. Note 2 appears twice in both the array.

Explanation 2:

Elements (2, 10) appears in both the array.

Q17. list of squares

WAP a Program to Generate a list of squares using list comprehensions.

Output 1

[1, 4, 9, 16, 25]

Q18. Filter even numbers

WAP a Program to Filter even numbers using list comprehensions.

Output 1

[2, 4, 6, 8, 10]

Q19. List of uppercase characters

WAP a Program to Create a using list comprehensions

Input 1:

text = "hello"

Output 1:

['H', 'E', 'L', 'L', 'O']

Q20.Find the output:-

```
a = 4
```

```
print(type(a))
```

```
a = 4,
```

```
print(type(a))
```

```
a = (4)
```

```
print(type(a))
```

```
a = ()
```

```
print(type(a))
```

```
a = (4,)
```

```
print(type(a))
```

```
a = (4,5)
```

```
print(type(a))
```